

CZ5101 Midterm Test (Closed book)

Friday, 3 October 2008

1. Which of the two alternative ways of machine computing in floating point arithmetic the area of a triangle given the lengths of its side a , b , and c , is more accurate? Explain why and in what case the formula becomes inaccurate.

a. Let $s = (a + b + c)/2$, $A = \sqrt{s(s-a)(s-b)(s-c)}$

b. Assuming $a \geq b \geq c$, $A = \sqrt{(a+(b+c))(c-(a-b))((c+(a-b))(a+(b-c)))/4}$

b is more accurate, read "What every computer scientist should know about floating-point arithmetic," page 182 & 183.

8+8 marks

2. (a) Find the LU decomposition of the following matrix. (b) And compute the determinant of the matrix using the result in (a).

$$\begin{bmatrix} 1 & 1 & 3 \\ 4 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

The factors are (without pivoting)

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 0 & -2 & -11 \\ 0 & 0 & -3 \end{bmatrix}$$

*The determinant is $1 * (-2) * (-3) = 6$.*

10 + 6 marks

3. Suppose we want to interpolate our data known at some discrete set of points by a ratio of polynomials as

$$\frac{a_0 + a_1x + a_2x^2}{1 + b_1x}.$$

- a. How many points do we need in order to determine the expression uniquely.
b. Give the explicit form of the unknown coefficients in terms of the data set (x_i, y_i) .

We need four points if the resulting equation is not singular. The linear equations determining the coefficients are

$$\begin{bmatrix} 1 & x_1 & x_1^2 & -x_1y_1 \\ 1 & x_2 & x_2^2 & -x_2y_2 \\ 1 & x_3 & x_3^2 & -x_3y_3 \\ 1 & x_4 & x_4^2 & -x_4y_4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ b_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

6 + 10 marks

4. Consider numerical integration.
- Give the expression T_N for the extended formula of trapezoidal rule of N points.
 - In the Romberg integration, how do we use T_N to extrapolate to $N \rightarrow \infty$.

$$T_N = h \left(\frac{1}{2} f_1 + f_2 + \dots + f_{N-1} + \frac{1}{2} f_N \right) + O\left(\frac{1}{N^2}\right)$$

We do polynomial extrapolation in variable h^2 .

8 + 8 marks

5. We want to generate random numbers in the interval $(0, \infty)$ with a probability density $p(x) \propto \exp(-kx)$, where k is some positive constant. This can be done with a transformation method. Give a formula for the random number x in terms of a uniformly distributed random number $\xi \in (0, 1)$.

The cumulative distribution function is

$$F(x) = \int_0^x c e^{-ky} dy = \frac{c}{k} (1 - e^{-kx}) = \xi$$

Since we need $F(+\infty) = 1$, we demand $c = k$. Thus, $x = -(1/k) \ln(1-\xi)$ or $-(1/k) \ln \xi$.

10 marks

6. Consider Monte Carlo method.
- State the “detailed balance” condition in Markov chain Monte Carlo.
 - Outline in pseudo-code form a program to simulate a one-dimensional Ising model of N sites with periodic boundary condition.
 - Discuss how the average energy and its associated Monte Carlo error can be computed.

The detailed balance condition is

$$P(X)W(X \rightarrow Y) = P(Y)W(Y \rightarrow X)$$

where X and Y are arbitrary states, $P(X)$ is probability of state X , and W is transition matrix.

set the initial spins $\sigma_i = 1$ (or at random) for $i = 0$ to $N-1$.

for MCSTEP = 1 to MAX {

*pick a site i at random by $i = N * \xi$ ($0 < \xi < 1$ is a random number)*

if $\Delta E = 2\sigma_i(\sigma_{(i-1) \bmod N} + \sigma_{(i+1) \bmod N}) \leq 0$, OR $\xi' < \exp(-\Delta E/(k_B T))$, flip the spin by $\sigma_i = -\sigma_i$

}

Average energy and energy square are calculated by sample average. But using variance to estimate the error for $\langle E \rangle$ under estimates the error due to sample correlation. A binning method is preferred for accurate estimate of the error.

6 + 10 + 10 marks